

Fifteenth Section. The form problem of the group G_{168} and the theory of equations of the seventh degree.

§ 125. The resolvents of the form problem.

For the group G_{168} , the form problem gives, according to the general theory, an equation of degree 168 whose coefficients are invariants. Each divisor of the group gives a resolvent of lower degree, namely of degree equal to the index of the divisor. We shall use here the two most interesting cases, the octahedral group

$$G_{24} = \langle \chi^2, \omega, \Theta^\nu \rangle, \quad (1)$$

and the group of degree 21

$$G_{21} = \langle \tau, \Theta\chi \rangle, \quad (2)$$

which lead to resolvents of the seventh and eighth degrees.

For the resolvents of the seventh degree one may start from the principal axes of the group. A principal axis remains fixed under the group G_{24} , and under the whole group is carried into three distinct lines. Thus there are seven triples of principal axes, determined by an equation of the seventh degree. With the notation of § 118 put

$$\begin{aligned} A_1 &= \alpha\gamma x_1 + \gamma\beta\varepsilon^3 x_2 + \alpha\beta\varepsilon^2 x_3, \\ A_2 &= \beta\gamma\varepsilon^3 x_1 + \alpha\beta\varepsilon^2 x_2 + \alpha\gamma x_3, \\ A_3 &= \alpha\beta\varepsilon^2 x_1 + \alpha\gamma x_2 + \beta\gamma\varepsilon^3 x_3. \end{aligned} \quad (3)$$

Then $A_1 = 0, A_2 = 0, A_3 = 0$ are the equations of three of these axes. The substitution χ carries A_1, A_2, A_3 cyclically into one another, and the other generating substitutions merely permute them. Therefore the symmetric functions of these three quantities are roots of resolvents of degree seven.

The simplest such symmetric function is

$$A_1^2 + A_2^2 + A_3^2. \quad (4)$$

Using the relations among α, β, γ one finds

$$\mu = \frac{2\varepsilon}{7}, \quad \lambda = -\varepsilon \frac{1 + \sqrt{-7}}{14}, \quad \frac{\mu}{\lambda} = -\frac{1 - \sqrt{-7}}{2}. \quad (5)$$

Thus, if $A_1^2 + A_2^2 + A_3^2 = \lambda z$,

$$z = x_1^2 + x_2^2 + x_3^2 - \frac{1 - \sqrt{-7}}{2}(x_2 x_3 + x_3 x_1 + x_1 x_2). \quad (6)$$

The other roots are obtained by the powers of τ :

$$\begin{aligned} z_r &= \varepsilon^{2r} x_1^2 + \varepsilon^{4r} x_2^2 + \varepsilon^r x_3^2 \\ &\quad - \frac{1 - \sqrt{-7}}{2}(\varepsilon^{-r} x_2 x_3 + \varepsilon^{-2r} x_3 x_1 + \varepsilon^{-4r} x_1 x_2), \quad r = 0, 1, \dots, 6. \end{aligned} \quad (7)$$

The coefficients of the equation whose roots are the seven quantities z_r are invariants. If a_ν is the coefficient of $z^{7-\nu}$, then a_ν is an invariant of degree 2ν . Hence $a_1 = 0$. The other coefficients are linear and homogeneous combinations of

$$\begin{array}{c|l} a_2 & f \\ a_3 & \Delta \\ a_4 & f^2 \\ a_5 & \Delta f \\ a_6 & \Delta^2, f^3 \\ a_7 & C, \Delta f^2. \end{array} \quad (8)$$

In the special case $f = 0$, comparison of leading terms in the power sums gives

$$\sum z_r^3 = -3 \cdot 7 \frac{1 - \sqrt{-7}}{2} x_1^5 x_2 + \dots, \quad (1)$$

$$\sum z_r^6 = \left[6 \cdot 7 + 15 \cdot 7 \left(\frac{1 - \sqrt{-7}}{2} \right)^2 \right] x_1^{10} x_2^2 + \dots. \quad (2)$$

Newton's formulae then give

$$z^7 - 7 \frac{1 - \sqrt{-7}}{2} \Delta z^4 - 7 \frac{5 + \sqrt{-7}}{2} \Delta^2 z - C = 0. \quad (9)$$

With

$$z = \frac{Cu}{\Delta^2}, \quad g = \frac{\Delta^7}{C^3}, \quad (10)$$

this becomes

$$u^7 - 7 \frac{1 - \sqrt{-7}}{2} gu^4 - 7 \frac{5 + \sqrt{-7}}{2} g^2 u - g^2 = 0. \quad (11)$$

In the general case another parameter is required,

$$h = \frac{f \Delta^4}{C^2}. \quad (12)$$

The coefficients of the general resolvent are, apart from numerical factors,

$$h, \quad g, \quad h^2, \quad hg, \quad (g^2, h^3), \quad (g^2, h^2g), \quad (3)$$

and direct comparison gives

$$\begin{aligned} & 7 \frac{1 - \sqrt{-7}}{2} h, \\ & -7 \frac{1 - \sqrt{-7}}{2} g, \\ & -7(4 + \sqrt{-7})h^2, \\ & 14(2 + \sqrt{-7})hg, \\ & -7 \frac{5 + \sqrt{-7}}{2} g^2 - 7 \frac{7 + 3\sqrt{-7}}{2} h^3, \\ & -g^2 + \frac{167 - 7\sqrt{-7}}{2} gh^2. \end{aligned} \quad (13)$$

The functions g, h are fractional invariants which depend only on the ratios $x_1 : x_2 : x_3$. Every fractional invariant with the same property is rationally expressible through g, h . Indeed, its numerator and denominator are whole invariants of the same degree, and therefore are rationally expressible in f, Δ, C . If a term has the form

$$f^a \Delta^b C^c, \quad (4)$$

then, for common degree m ,

$$4a + 6b + 14c = m. \quad (14)$$

§ 126. Reduction of the general resolvent of the seventh degree to the special one.

For a point (x) write

$$u_x = \frac{\Delta_x^2}{C_x}, \quad g_x = \frac{\Delta_x^7}{C_x^3}, \quad h_x = \frac{f_x \Delta_x^4}{C_x^2}. \quad (1)$$

The general resolvent is

$$R(u_x, g_x, h_x) = R_x = 0, \quad (2)$$

and the special resolvent is obtained by setting $f_x = 0$.

According to Klein, the solution of the general resolvent can be reduced to that of the special resolvent if one adjoins a root of a biquadratic equation. Introduce a second point (y) , and form the polar of f with respect to (x) :

$$f_1(x, y) = y_1 f_{x_1}(x) + y_2 f_{x_2}(x) + y_3 f_{x_3}(x). \quad (3)$$

Require (y) to lie both on the ground curve and on this polar,

$$f(y) = 0, \quad f_1(x, y) = 0. \quad (4)$$

Then to a point (x) there correspond four points (y) . When (x) is replaced by a connected point under the group, these four points also go into connected points. For such a point (y) , one has $h_y = 0$; the quantity g_y , however, is a four-valued function of (x) . Its symmetric functions are invariants of G_{168} , and hence g_y satisfies a biquadratic equation with coefficients rational in g_x, h_x .

Substituting the corresponding value u_y into the special resolvent gives

$$\Phi(u_y, u_x, g_x, h_x) = 0. \quad (5)$$

If (x) is chosen so that no two of the four points y', y'', y''', y'''' are connected, then $\Phi = 0$, considered as an equation in u_x , has only one root in common with $R_x = 0$. The greatest common divisor of R and Φ therefore gives u_x rationally in terms of u_y, g_x, h_x . This is the desired reduction.

The same group contains a divisor of degree 21, generated by τ and χ , which gives a resolvent of degree eight. One may take $x_1 x_2 x_3$ as a root. Its coefficients are invariants rising to degree 94.

§ 127. The permutation group of degree 168.

The group of the equations obtained above is the group of ternary linear congruence substitutions modulo 2. The seven non-zero triples are

$$(1, 0, 0), (0, 1, 0), (0, 0, 1), (0, 1, 1), (1, 0, 1), (1, 1, 0), (1, 1, 1). \quad (1)$$

A substitution

$$\begin{pmatrix} a & b & c \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix} \quad (2)$$

with entries taken modulo 2 permutes these seven triples, and the totality of these permutations is P_{168} . With

$$\tau = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad (3)$$

and with $z_0 = (1, 0, 0)$, the successive powers of τ arrange the seven symbols cyclically. The two generators are even permutations, and therefore P_{168} is a subgroup of the alternating group on seven letters.

§ 128. Equations of the seventh degree with a group of degree 168.

Let $\lambda_0, \lambda_1, \dots, \lambda_6$ be seven quantities and let

$$\tau = (\lambda_0, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6) \quad (1)$$

be a cycle of degree seven. Add

$$\omega = (\lambda_1, \lambda_2)(\lambda_3, \lambda_5). \quad (2)$$

These permutations generate P_{168} . In this notation

$$\Theta^3 = \tau^6 \omega \tau \omega = (1, 2, 5, 3)(4, 6), \quad \Theta = (1, 3, 5, 2)(4, 6), \quad (5)$$

$$\chi = \omega \tau^3 \Theta \tau^2 = (1, 4, 2)(3, 5, 6). \quad (6)$$

The element λ_0 is fixed by the octahedral subgroup P_{24} . A function belonging to P_{168} is obtained by taking

$$\begin{aligned} v = & \lambda_0 \lambda_2 \lambda_6 + \lambda_1 \lambda_5 \lambda_0 + \lambda_2 \lambda_6 \lambda_1 + \lambda_3 \lambda_0 \lambda_2 \\ & + \lambda_4 \lambda_1 \lambda_3 + \lambda_5 \lambda_2 \lambda_4 + \lambda_6 \lambda_3 \lambda_5. \end{aligned} \quad (3)$$

This remains unchanged under ω and τ , hence under all of P_{168} . Consequently v is a root of a degree-thirty equation with symmetric coefficients; after adjoining the square-root of the discriminant it splits into two factors of degree fifteen.

If v belongs to the rationality domain, together with the symmetric functions of the λ , then the λ 's are roots of a special equation of the seventh degree whose Galois group has order 168. It remains to form three rational functions X_1, X_2, X_3 of the roots which transform according to the corresponding ternary substitutions of G_{168} . Substituting these functions in the invariants of G_{168} then gives quantities in the rationality domain, and any function of X_1, X_2, X_3 changed by G_{168} is a root of a total resolvent.

§ 129. Contragredient groups.

If

$$y = A(x), \quad \eta = A^{-t}(\xi), \quad (1)$$

are transposed substitutions, then the rows (x_1, x_2, x_3) and (ξ_1, ξ_2, ξ_3) are contragredient. For derivatives this gives the rule

$$\xi_1^\kappa \xi_2^\lambda \xi_3^\mu = \sum C_{\alpha\beta\gamma}^{\kappa\lambda\mu} \eta_1^\alpha \eta_2^\beta \eta_3^\gamma, \quad \kappa + \lambda + \mu = m, \quad (2)$$

and hence

$$\frac{\partial^m \varphi}{\partial x_1^\kappa \partial x_2^\lambda \partial x_3^\mu} = \sum C_{\alpha\beta\gamma}^{\kappa\lambda\mu} \frac{\partial^m \varphi}{\partial y_1^\alpha \partial y_2^\beta \partial y_3^\gamma}. \quad (3)$$

This proves the transformation law for higher derivatives. Therefore powers and products of variables in a contravariant may be replaced by the corresponding differential operators applied to a covariant, and conversely. If A runs through a group G , then A^{-t} runs through the contragredient group. The invariants of the latter are the contravariants of G .

§ 130. Solution of the seventh-degree equation with group P_{168} by the form problem of G_{168} .

The group G_{168} is contragredient to itself. Its generators remain in the group under transposition. Hence its contravariants coincide with its invariants, up to notation. The octahedral subgroup G_{24} , however, is not equal to its contragredient.

Let η_0 be a function of the roots $\lambda_0, \dots, \lambda_6$ belonging to P_{24} , for instance a rational function of λ_0 , and let $\eta_0, \eta_1, \dots, \eta_6$ be its cyclic conjugates. With the forms

$$z_r = \varepsilon^{2r} x_1^2 + \varepsilon^{4r} x_2^2 + \varepsilon^r x_3^2 - \frac{1 - \sqrt{-7}}{2} (\varepsilon^{-r} x_2 x_3 + \varepsilon^{-2r} x_3 x_1 + \varepsilon^{-4r} x_1 x_2) \quad (1)$$

put

$$\psi = \sum_{r=0}^6 \eta_r z_r. \quad (2)$$

Then

$$\psi = p_1x_1^2 + p_2x_2^2 + p_3x_3^2 + 2q_1x_2x_3 + 2q_2x_3x_1 + 2q_3x_1x_2. \quad (3)$$

Taking three such functions gives three forms ψ, ψ', ψ'' . Their functional determinant

$$\Psi = \begin{vmatrix} \psi_{x_1} & \psi_{x_2} & \psi_{x_3} \\ \psi'_{x_1} & \psi'_{x_2} & \psi'_{x_3} \\ \psi''_{x_1} & \psi''_{x_2} & \psi''_{x_3} \end{vmatrix} \quad (4)$$

is a cubic form

$$\Psi = \sum A_{ijk}x_ix_jx_k. \quad (5)$$

Its coefficients are trilinear functions of the three systems $\eta_r, \eta'_r, \eta''_r$. By using contragredient variables and then applying the theorem on substituting derivatives in invariants, one obtains three functions

$$X_1, \quad X_2, \quad X_3, \quad (6)$$

which undergo exactly the linear substitutions corresponding to the permutations of P_{168} . This is the required connection with the form problem.

§ 131. Possibility of determining the functions X_1, X_2, X_3 .

It remains to show that $\eta_0, \eta'_0, \eta''_0$ can be chosen so that X_1, X_2, X_3 are not identically zero. From the calculation of the preceding paragraph one obtains expressions of the following type:

$$\begin{aligned} X_1 &= (p_3, p'_2, q''_1) + (q_1, p'_1, q''_2) + (p_1, q'_3, p''_3), \\ X_2 &= (q_2, q'_1, p''_3) + (p_3, p'_2, q''_2) + (q_3, p'_2, q''_1), \\ X_3 &= (q_2, q'_3, q''_2) + (p_3, q'_1, p''_3) + (q_2, p'_2, p''_3), \end{aligned} \quad (1)$$

where the bracket denotes the determinant formed from the indicated rows of coefficients. These functions are trilinear in the three chosen systems.

Make the substitution

$$\eta_r = \sum_s a_s \lambda_r^s, \quad \eta'_r = \sum_s a'_s \lambda_r^s, \quad \eta''_r = \sum_s a''_s \lambda_r^s. \quad (2)$$

The determinant of this transformation is, up to sign, the product of all differences $\lambda_i - \lambda_k$, and is non-zero when the roots are distinct. Hence rational values of a_s, a'_s, a''_s can be chosen so that the three trilinear forms do not vanish. The required functions X_1, X_2, X_3 therefore exist, and the reduction of the seventh-degree equation with group P_{168} to the form problem of G_{168} is complete.